



HimBhasha AI: Preserving and Empowering Himachal's Regional Languages through Artificial Intelligence

Track: Regional Language AI (Voice, Translation, Low-Resource NLP)

Problem Statement

Himachal Pradesh is home to a rich tapestry of regional languages, including Kangdi, Mandeali, Kullui, Chambeali, Sirmauri, and Kinnauri. Unfortunately, these languages are severely underserved by current AI technologies. Digital assistants primarily support English and Hindi, creating significant barriers for native speakers in accessing essential services such as government portals, healthcare information, educational resources, and general digital platforms. This digital divide not only hinders daily life but also poses a critical threat to the preservation of these invaluable cultural linguistic assets.

Proposed Solution: HimBhasha AI

HimBhasha AI is envisioned as an intelligent, multilingual assistant designed to bridge this linguistic gap and empower the people of Himachal Pradesh. Our initial Minimum Viable Product (MVP) will focus on Kangdi, with a robust architecture designed for seamless expansion to other regional languages. The core features will include:

- **Text and Voice Communication:** Interact naturally in your preferred regional dialect.
- **Multilingual Translation:** Effortlessly translate between Kangdi, other Himachal regional languages, Hindi, and English.
- **Document OCR & Simplification:** Upload documents, extract text using OCR, and receive simplified explanations in your chosen language.
- **AI Question Answering:** Get accurate information and answers tailored to your regional context.
- **Natural Voice Responses:** Receive information in a clear, natural-sounding voice.

Key Features

-  **Voice-to-Voice AI Assistant:** Natural, conversational AI interaction.
-  **Multilingual Translation:** Seamless language conversion.

-  **Document OCR & Simplification:** Extract and understand document content easily.
-  **AI Question Answering:** Context-aware information retrieval.
-  **Regional Language Learning:** Support for learning and practicing local dialects.
-  **Government Service Assistant:** Simplified access to public services.
-  **Offline-ready Architecture:** Ensuring accessibility even with limited connectivity.

AI Workflow

AI Workflow Flowchart

User Input (Voice/Image/Text) → Speech Recognition/OCR → Language Detection → Translation + AI Understanding → AI Response Generation → Text + Voice Output

Technology Stack & Roadmap

Tech Stack

Our technology stack is built for scalability, performance, and rapid development:

Frontend:

- React
- Next.js
- Tailwind CSS

Backend:

- FastAPI

AI Models:

- Whisper (Speech Recognition)
- OCR (PaddleOCR / Tesseract)
- Gemini API (LLM Capabilities)
- Indic NLP Models
- Text-to-Speech (TTS)

Database:

- Firebase / Supabase

Deployment:

- Vercel / Render

These components are represented by modern, rounded cards for a clean, technology-focused aesthetic.

This document is designed for a modern, clean, and minimal aesthetic, with subtle Indian-inspired elements. The color palette includes deep blue, white, saffron, and green, utilized with soft gradients. Visuals such as AI-themed illustrations, language icons, voice wave graphics, and a map outline of Himachal Pradesh will be incorporated to create an infographic-style presentation suitable for a national hackathon submission in A4 PDF format.

Expected Impact

- **Enhanced Digital Accessibility:** Significantly improve access to digital services for speakers of Himachal's regional languages.
- **Bridged Communication Gap:** Foster better communication between local communities and government, healthcare, and educational institutions.
- **Language Preservation:** Utilize AI to document, digitize, and revitalize low-resource regional languages.
- **Scalable Multilingual Platform:** Build a robust foundation that can be expanded to support other low-resource Indian languages.

Future Roadmap

- **Phase 1 (MVP):** Focus on **Kangdi** language support.
- **Phase 2:** Expand to **Mandeali** and **Kullui**.
- **Phase 3:** Integrate **Chambeali**, **Sirmauri**, and **Kinnauri**.
- **Phase 4:** Generalize and adapt the platform for other **low-resource Indian languages**.